Management Science and Engineering Department

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Simulation Modeling and Analysis

Confetti Production Facility Detailed Report

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1. Introduction & Problem Description

The system under study is a confetti tube production line that manufactures three types of tubes:

- Small
- Large
- Extra Large (XL)

Tubes pass through a series of processes that include paper body rolling, trimming, several chemical operations, body preparation, CNC cutting, gluing & drying, final assembly, and finally labeling & packaging. Finished tubes are stored in an acceptance / finished-goods area.

Management is concerned that the current system

- wastes time due to congestion and blocking,
- fails to meet daily demand,
- may not be using raw materials efficiently, and
- generates excessive work-in-process (WIP) inventory.

The purpose of this project is to build, verify, and validate a discrete-event simulation model of the system using Simio; then use the model to (1) understand current performance and main bottlenecks and (2) make recommendations for improvement. This report describes the assumptions, model structure, verification/validation activities, and key results from the latest fixed version of the model.

2. Data, Inputs, and Modeling Assumptions

2.1. Seasonal Demand and Arrivals

Demand varies by season. A base demand of

- **DailyDemand** = 200 tubes per day

defined in the experiment for each scenario:

- **Fall**: 200 tubes/day
- **Spring**: 300 tubes/day
- **Winter**: 40 tubes/day

The model runs for **RunDays = 60** days in each scenario.

Arrivals are generated by a single **Source** object TubeArrivals using the following logic:

- Each created entity has a property TubeType (Small, Large, XLarge) drawn from a table TubeData using the column TubeMixWeight (random row selection).
- Interarrival time is defined by the expression

ShiftMinutes / DailyDemand

where ShiftMinutes is the number of working minutes per day (see below). This yields the right **expected number of arrivals per day** for each season.

This structure allows the experiment scenarios to change seasonal volume simply by modifying SeasonRateFactor, without changing the base model logic.

2.2. Shift Structure and Run Length

The system operates on a single shift with the following model-level properties:

- ShiftMinutes = 480 minutes (8 hours per day)
- RunDays = 60 days

The work schedule element DailyWorkSchedule defines the working periods:

- 7:00 AM – 12:00 PM → 5 hours
- 1:00 PM – 4:00 PM → 3 hours

Non-working time is treated as off-shift; resources are unavailable during these periods. The experiment warm-up period is **0 days,** and each scenario uses multiple replications (15 in the current version) to reduce stochastic noise.

2.3. Process Flow and Service Times

All tubes pass through the same ordered sequence of stations:

1. Rolling
2. Trimming
3. ChemicalPressing
4. ChemicalMixing

5. BodyPreparation
6. CNC
7. GlueAndDrying
8. FinalAssembly
9. LabelingAndPackaging
10. FinishedGoods sink

Routing is implemented with a **Process Sequence Table**, which captures both the route and processing times. The table includes the columns

- TubeType (Small, Large, XLarge)
- ProcessSequence (Input@Rolling, Input@Trimming, …, Input@FinishedGoods)
- ServiceTimeExpression (Simio expression for processing time in minutes)

The servers use the following configuration:

- Capacity Type = Fixed
- Initial Capacity = (ServerName)Capacity
- Process Type = Specific Time
- Processing Time = ProcessSequenceTable.ServiceTimeExpression

Service times are modeled primarily with **Triangular distributions** estimated from the instructional videos and problem description. Typical examples:

- Small tubes at some stations: Random.Triangular(1.5, 2, 4) minutes
- Large tubes at some stations: Random.Triangular(2.5, 4, 5) minutes
- XL tubes generally require longer processing times

Exact parameters differ by station and tube type; the table-based approach ensures all stations use a consistent and easily modifiable data source.

2.4. Raw Materials and Inventory Assumptions

The system consumes several types of raw materials. These were modeled with **Material elements** and **Timer-triggered restocking processes**.

2.4.1. Materials

Defined Material elements:

- PaperTubes
- Chemicals
- HardwareUnits
- PVCBoards
- AssemblyFixtures

Each material has Location Based Inventory = False and an initial quantity of 0; all inventory is generated by restocking processes.

2.4.2. Restocking Logic

Five **Timer elements** generate periodic restocking events:

- Paper_QuarterlyTimer with 12-week interval
- Chem_WeeklyTimer with 1-week interval
- Hardware_WeeklyTimer with 1-week interval
- PVC_MonthlyTimer with a 4-week interval
- Fixtures_BiMonthlyTimer with an 8-week interval

Each timer triggers a corresponding process (Restock_Paper, Restock_Chem, etc.) that uses Produce steps to add materials according to the problem description:

- Paper: lot size 5000 units (5 rolls × 1000 tubes per roll)
- Chemicals: 500 units per delivery
- Hardware units: 300 assemblies per delivery
- PVC boards: 400 mounts per delivery
- Assembly fixtures: 1000 units per delivery

This structure ensures that material shortages do not dominate system performance over the 60-day horizon, consistent with the narrative that **capacity**, not materials, is the main constraint.

2.4.3. Material Consumption

Consumption occurs in the **add-on process triggers** for the relevant stations. For example: In ChemicalMixing, an add-on process, Consume_Chem is executed **Before Processing** and contains a Consume step:

- Material Name = Chemicals
- Quantity = 1

Similar logic is used for other materials in GlueAndDrying, FinalAssembly, etc. The model assumes one unit of material per entity at the appropriate step.

2.5. States and Custom Statistics

To study congestion and blocking, additional model-level states and statistics are defined.

2.5.1. Model and Entity States

- Model state BlockedCount (Real, initialized to 0): that counts the total number of blocking events.

- Entity state BlockStart: stores the time when the entity begins being blocked (which is 0 if not currently blocked).

2.5.2. Tally Statistic

- Tally statistic element BlockedWaitTime with unit type **Time**.

This tally records the duration of each blocking incident experienced by an entity.

3. The Model Structure Overview

Our Final Simio model consists of the following:

- **Facility view** that shows the main flow line as well as all servers in sequence (correct order).
- **Data Tables** (TubeData, ProcessSequenceTable) to control the mix and service times.
- **Elements** for materials, work schedule, timers, and tallies.
- **Processes** for material restocking and blocking statistics.
- **Experiment** with scenarios for Fall, Spring, and Winter seasonality.

This separation of logic and data makes the model relatively easy to maintain; changing service times, product mix, or raw material plans can be done via tables and properties without editing the core process logic.

4. Verification and Validation

4.1. Verification – Ensuring the Model Was Built Correctly

Verification concentrates on checking whether the implemented Simio model reflects the conceptual logic we wanted to present. We performed several structural checks with the help of the recommended practices from the Simio references to confirm that the model is correct.

4.1.1. Flow Tracing and Animation

The full flow of entities was tracked through the line using Simio's step-through mode. We observed entities as they moved server by server, and we verified that each tube followed its sequence and that no errors like unintentional loops, missing links, or routing mistakes existed. During this process, some issues were presented, and we fixed them, making sure that the final flow matched the model logic and the ProcessSequenceTable.

4.1.2. Routing and Processing Time Verification

To prevent hard-coded parameters and to ensure consistency, the processing time for each server was checked to verify it matched the ProcessSequenceTable.ServiceTime Expression.

4.1.3. Arrival Logic Verification

The number of entities created per day was confirmed by having some test runs, which also matched the theoretical value (DailyDemand x SeasonRateFactor). In this way, the interarrival time expression validated that it produces the expected capacity for each scenario.

4.1.4. Warning Resolution: Excessive Entity Creation

In the early stages of testing, a warning of "Objects created are more than Simio's Limit" indicated that entities were being created in excess of the default internal maximum. As a result, we increased the allowed maximum entity limit after confirming that it wasn't a model error but rather the high WIP levels characteristic of the system. This eliminated the warning and allowed entities to be created without any constraints.

4.1.5. Material Restocking and Consumption Verification

To certify that restocking events happened at the proper intervals (weekly, monthly, quarterly), timers were checked. Also, one unit of material per entity was subtracted at the suitable stations when consumption was verified.

4.1.6. Blocking Detection Logic Verification

We used the blocking conditions to reduce downstream buffer capacities, which guaranteed accurate congestion tracking. This allowed the following to be confirmed:

- BlockedCount increased anytime downstream stations were full.
- BlockedWaitTime recorded precise waiting times.
- BlockStart resets once the object starts moving.

4.1.7. Replication and Statistical Verification

Our results were statistical as the model was tested using 15 replications. This reduced sampling error and produced stable confidence intervals.

4.2. Validation – Ensuring the Model Represents the Real System

Validation measures how well the verified model performs in comparison to expectations from reality.

4.2.1. Face Validity with Stakeholder Knowledge

The behavior of the system was a clear reflection of recognized operational problems, such as WIP accumulation upstream, prolonged processing times and delays, and high traffic at downstream stations. This supported management's claim that significant blocking and bottlenecks invade the actual system.

4.2.2. Bottleneck Identification

The main bottleneck of the simulation was LabelingAndPackaging, followed by FinalAssembly and CNC as minor limitations. It is consistent with a timed flow line in which quantity is controlled by downstream stations.

4.2.3. Material Behavior Validation

During the 60-day run, there were no material shortages since material replacement capacities were substantial in relation to production output. This confirms that the constraints of the model accurately represent reality by matching the assumption that the availability of material isn't the limiting factor.

4.2.4. Comparison with Theoretical Principles

The Confetti Tube system is complex; therefore, we could not use the M/M/1 analytical queueing formula to compare. However, the model displayed some queueing behavior like rising arrival rates leading to a super-linear increase in WIP and system time.

5. Experiment Design and Performance Measures

5.1. Experiment Scenarios

The main experiment, which is experiment 1, includes three implemented scenarios:

1. Fall (scenario 1)
    - DailyDemand = 200
2. Spring (scenario 2)
    - DailyDemand = 300

3. Winter (scenario 3)
    - DailyDemand = 40

Each scenario uses:

- **Run length:** 60 days
- **Warm-up:** 0 hours
- **Replications:** 15
- **Confidence level:** 95%

5.2. Key Responses

The following responses are of primary interest:

1. ThroughputPerDay
    - Expression: FinishedGoods.InputBuffer.NumberEntered / RunDays
    - Measures the average number of tubes completed per day.
2. Station utilizations (each server)
    - Data source: Server
    - Category: Capacity
    - Data item: ScheduledUtilization
3. Time in system and WIP
    - For each TubeType and overall ModelEntity population:
        - TimeInSystem (Average)
        - NumberInSystem (Average)

These responses allow us to assess whether the system can meet demand and where congestion is worst.

6. Results-Current System Performance

6.1. Throughput vs. Demand

Across all three seasons, the simulation shows that:

- Average throughput is roughly 20 tubes per day.
- Throughput is almost independent of the demand level, indicating a strong capacity bottleneck.

Comparison with demand:

- Fall: Demand ~ 200/day; output ~ 20/day, which is only about 10% of demand is met.
- Spring: Demand ~300/day; output ~20/day, which is an even smaller percentage of demand met.
- Winter: Demand ~ 40/day; output ~20/day, which is roughly 50% of demand is met.

The system is clearly incapable of satisfying demand in any season within the current configuration.

6.2. Time in System and Work-in-Process

From the ModelEntity population statistics:

- The average TimeInSystem is extremely high, ranging from approximately 300 to 400 hours across different entity types and scenarios (Fall, Spring, Winter), which is equivalent to 25–28 days, often exceeding several weeks.

    - The maximum TimeInSystem values are between 1,245 and 1,265 hours for some entities, which is about 52 days. This shows that certain products stay in the system for almost the full simulation period.

    - The average NumberInSystem is high with:

        - 203 units for large entities
        - 366 units for small entities

- 268 units for XL entities

This shows that on average, several hundred tubes are in the system at all times, which indicates extreme congestion and an increase in WIP.

6.3. Resource Utilizations

Server utilization statistics show that:

- In every scenario (fall, spring, and winter), ChemicalMixing is at 100% ScheduledUtilization, which means that it is active and completely utilised constantly.
- GlueandDrying in the fall is 99.1%, 99.1% in the spring, and 96.3% in the winter, which is very high utilisation. This indicates near-capacity operation and queuing.
- CNC utilisation reaches 96.1% in the fall, 95.7% in the spring, and 94.1% in the winter, which indicates that it is substantially loaded, especially during strong demand periods.
- Upstream stations (Rolling and Trimming) show moderate utilisation in the range of 80-85%, which indicates that they are busy but not experiencing significant limitations.

The primary bottleneck is ChemicalMixing, with secondary bottlenecks at GlueandDrying and CNC.

6.4. Raw Materials

Over a period of 60 days, the inventory of raw materials is monitored. During the run, with the selected lot sizes and delivery intervals, no material gets to a critically short level. This indicates that the throughput is limited by the processing capability instead of the material supply in the modeled time frame.

7. Interpretation and Identified Problems

From the experimental results, several main issues are identified.

1. **Severe Capacity Shortfall**
    - The line produces only about 20 tubes per day while demand ranges from 40 to 300 per day.
    - Demand in Fall and Spring is an order of magnitude larger than what the system can handle.
2. **Downstream Bottlenecks**
    - ChemicalMixing is our primary bottleneck with its utilization approaching 100% in all scenarios.
    - GlueandDrying and CNC also illustrate high utilization, which strengthens that the end of the line is the limiting segment.
3. **Excessive WIP and Lead Times**
    - Hundreds of tubes remain in process at any time, with average lead times measured in weeks.
    - This implies both high holding costs and very poor responsiveness to customer orders.
4. **Extensive Blocking Upstream**
    - Because downstream buffers are filled, upstream machines finish tasks but are unable to transfer entities, meaning wasting productive time.
5. **Raw Materials Not a Constraint**
    - Given the large lot sizes and restocking frequencies, raw materials are always available.
    - Investing more in raw material procurement would not improve throughput under the current capacity configuration.

Overall, the system behaves like a classic bottlenecked flow line where one or two stations at the end of the process dictate overall performance, while upstream resources spend considerable time blocked.

Recommendations and What-If Insights (Conceptual)

Even though our report looks into the current system only, we still found some areas to improve in the model and how the application might affect the model.

8.1. Increase Capacity at Chemical Mixing

This station is the main bottleneck in the system. In the model, this station always limits the production capability regardless of the demand fluctuation. So by improving the capacity of this station, either by adding another server or by enhancing its efficiency. It will directly lead to great improvement in the system output and also reduce the congestion upstream.

8.2. Increase Capacity at Glue and Drying (and Possibly CNC)

The Glue and Drying or the CNC station is expected to become the new constraints or bottlenecks of the model once we improve the Glue and Drying station. By increasing the capacity at these downstreams together with the improvements we made to the Glue and Drying station, it will help to maintain a balanced flow and also prevent new bottlenecks from forming.

8.3. Evaluate Additional Shift Time or Overtime

Another recommendation we thought about is increasing the daily working time to more than 480 minutes. For example, to 540 or 600 minutes. This will provide more processing time across the whole system. However, without the control of the capacity limitations, the longer the shifts can lead to the overloaded bottlenecks continuing to run without providing any real improvement to the system.

8.4. WIP Control and Buffer Management

The work-in-progress levels in our model are very high, which shows that our current buffer sizes can be larger than necessary. Implementing a WIP or Pull system, like the CONWIP or the Kanban, could help us in reducing the time taken in the system and also decrease the unnecessary accumulation of inventory. Even though the overall production will still be determined based on the system's available capacity.

8.5. Use the Model for Further Scenario Analysis

The simulation model is flexible and applies additional scenarios, for example, adjusting capacities, buffer limits, and shift lengths. This helps explore different trade-offs between the cost, flowtime, and throughput, and it helps us have a deeper understanding of how future improvements may affect the system performance.

Limitations and Future Work

The study has several limitations that should be considered when interpreting results:

9.1. Service Time Data Quality

For the processing times in the model, we have used the triangular distributions that were estimated after watching instructional videos for each step in the process, rather than detailed time studies. Due to this, the variability in the real-life operations might be different from the values used in the simulation.

9.2. Simplified Workforce and Scheduling

This model assumes a fixed daily schedule and does not include detailed elements such as break times, setup activities, and operator sharing between stations. These simplifications may affect the accuracy of the system's performance representation.

9.3. Simplified Demand Representation

For each season, the demand was modeled as a constant arrival rate instead of using actual customer orders or real-life ordering patterns for different seasons. Hence, as a result of that, the short-term fluctuations in the demand are not fully captured in this model.

9.4. Cost Analysis Not Fully Implemented

The model tracks operational measures, such as WIP and blocking, and includes financial factors like labor costs, equipment costs, and inventory holding costs, as well as lost sales. If we

consider these factors, it would enable us to conduct a more accurate economic assessment of the system.

9.5. No Automated Optimization

The main goal of this report was to have a better understanding of the current system and to identify any areas of improvement. An optimization study could be done by using tools like the OptQuest could be done in the future to give more detailed recommendations in regards to resource levels and scheduling decisions, and capacity planning.

Future work could address these limitations by collecting better data, enriching the schedule and cost structures, and performing optimization-based experiments to quantify the best combination of capacity additions and scheduling policies.

Conclusion

The detailed model of the confetti tube production system was developed, then we verified the model, and after that, it was qualitatively validated to make sure that it illustrates the right workflow. The seasonal demand experiments that we conducted showed that the current system cannot meet production requirements in any season. It achieves only about 20 tubes per day while the demand ranges from 40 to 300 tubes per day. The main bottleneck in the system was the Labeling and Packaging station, and the secondary constraints were the Final Assembly and CNC. The system shows that there are high levels of work in process and also a Long time in the system. Also, the analysis confirms that the raw materials do not limit throughput there with a period of sixty days.

Our findings suggest that the improvement efforts should focus on increasing the capacity in the downstream stations and the possibility of increasing the time of the daily shift, instead of investing additional resources in the raw materials. Even without doing a full optimization study, the simulation still works as an effective tool to evaluate any potential improvements that can be made to the confetti production process.